```
import numpy as np
import pandas as pd
df=pd.read_csv('outlier_data.csv')
print(df.head())
print(df.describe())


salary=df['Salary'].values
print(np.percentile(salary,25))
print(np.percentile(salary,50))
print(np.percentile(salary,75))


def outDetection(array):
    Q1,Q3=np.percentile(array,[25,75])
    IQR=Q3-Q1
    lr=Q1-(1.5*IQR)
    ur=Q3+(1.5*IQR)
    return lr,ur

lr,ur=outDetection(salary)
print("lower range =",lr,"upper range =",ur)

import seaborn as sns
import matplotlib.pyplot as plt
sns.boxplot(x=df['Salary'])
plt.show()

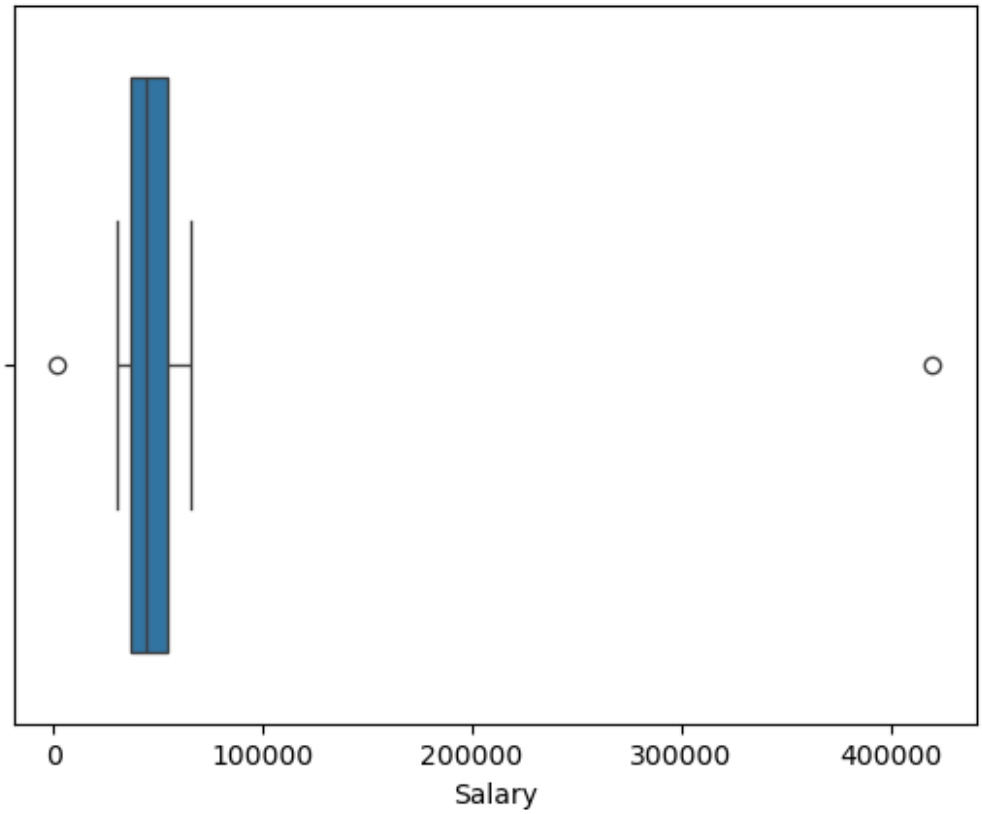

print(df[(df.Salary<lr)|(df.Salary>ur)])

new_df=df[(df.Salary>lr)&(df.Salary<ur)]
print(new_df)

lr1,ur1=outDetection(new_df.Salary.values)
print(lr1,ur1)
final_df=new_df[(new_df.Salary>lr1)&(new_df.Salary<ur1)]
print(final_df)
sns.boxplot(x=final_df['Salary'])
plt.show()
```

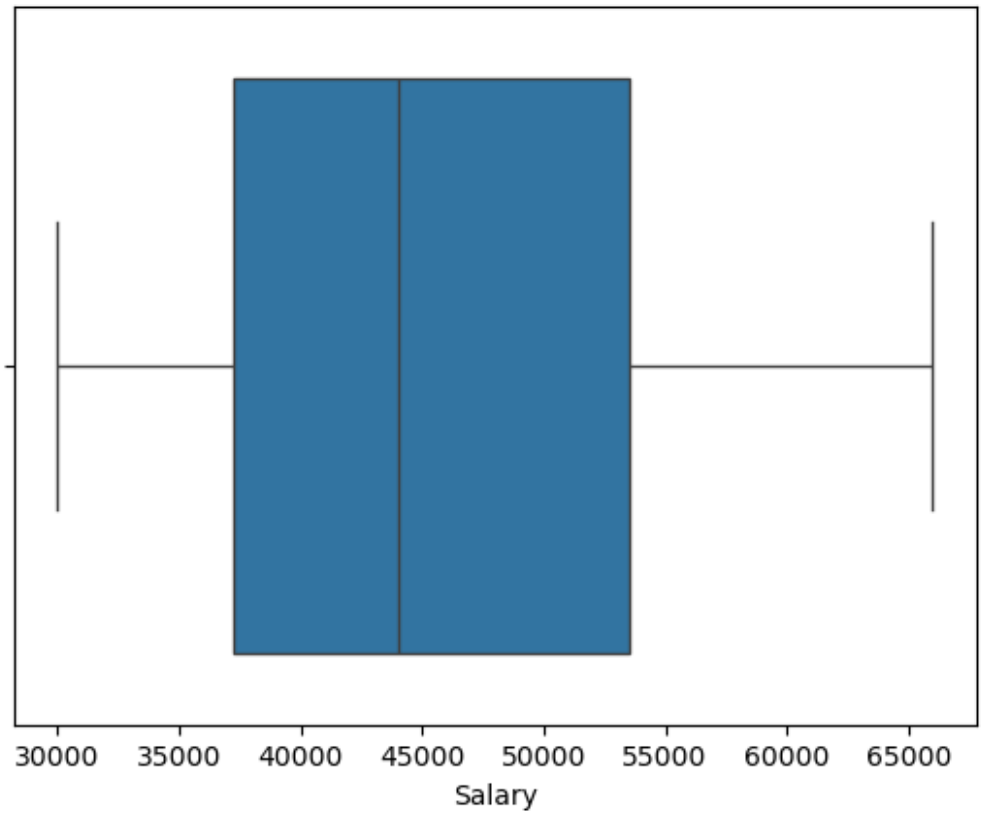
```
  Employee  Age  Experience  Salary
0      E01   25           2   32000
1      E02   28           4   38000
2      E03   30           6   42000
3      E04   27           3   35000
4      E05   35          10   55000
             Age  Experience         Salary
count  24.000000   24.000000      24.000000
mean   32.000000    6.541667   59333.333333
```

```
std     6.303898    3.956054   77956.323706
min    24.000000    1.000000    2000.000000
25%    27.750000    3.000000   36750.000000
50%    30.500000    6.000000   44000.000000
75%    35.000000    9.250000   54250.000000
max    52.000000   15.000000  420000.000000
36750.0
44000.0
54250.0
lower range = 10500.0 upper range = 80500.0
```




```
   Employee  Age  Experience  Salary
19      E20   52           3  420000
22      E23   26           2    2000
   Employee  Age  Experience  Salary
0       E01   25           2   32000
1       E02   28           4   38000
2       E03   30           6   42000
3       E04   27           3   35000
4       E05   35          10   55000
5       E06   32           8   48000
6       E07   29           5   40000
7       E08   41          15   66000
```

```
8        E09   38           12   58000
9        E10   26            2   33000
10       E11   34            9   52000
11       E12   31            7   45000
12       E13   36           11   57000
13       E14   24            1   30000
14       E15   33            8   50000
15       E16   29            5   41000
16       E17   40           14   63000
17       E18   27            3   36000
18       E19   30            6   43000
20       E21   28            4   37000
21       E22   35           10   54000
23       E24   32            7   47000
12875.0 77875.0
   Employee  Age  Experience  Salary
0        E01   25            2   32000
1        E02   28            4   38000
2        E03   30            6   42000
3        E04   27            3   35000
4        E05   35           10   55000
5        E06   32            8   48000
6        E07   29            5   40000
7        E08   41           15   66000
8        E09   38           12   58000
9        E10   26            2   33000
10       E11   34            9   52000
11       E12   31            7   45000
12       E13   36           11   57000
13       E14   24            1   30000
14       E15   33            8   50000
15       E16   29            5   41000
16       E17   40           14   63000
17       E18   27            3   36000
18       E19   30            6   43000
20       E21   28            4   37000
21       E22   35           10   54000
23       E24   32            7   47000
```


30000
35000
40000
45000
50000
55000
60000
65000
Salary